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Practical No. 1

Aim :- Perform installation and configuration of rasberry pios and familiarize with rasberry pi components & interfaces

--> Requirement

- Rasberry pi board
- Micro SD card (at 16GB recommended)
- Computer laptop
- Rasberry pi os.
- Rasberry pi image software
- Monitor
- HDMIT cable.
- Keyboard.
- Mouse
- Power adapter
- Internet connection.

--> Step to performs.

Step1:- Install Rasberry Pi imager.

Step2:- Select Rasberry Pi

Step3:- Select Rasberry Pi OS.

Step4:- Write the OS.

Step5:- Connect the Rasberry Pi

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Step 6:- Start Rasberry Pi

Step 7:- Configure the rasberry pi after installation check the Rasberry pi setting and make neccessary configuration such as Wi-Fi display keyboard and system updates.

Practical No. 2.

Aim: Perform a practical to connect Rasberry Pi with ethernet monitor, keyboard, mouse and USB device.

→ To connect Rasberry pi with ethernet, monitor, keyboard mouse USB device and varify their working

→ Requirement:

- Rasberry Pi
- Rasberry Pi OS installed Micro SD card
- Monitor
- HDMI cable.
- keyboard
- Mouse
- Ethernet Ca
- USB device (Pen drive)
- Power adapter

→ Ethernet :- Is ethernet a wired networking Technology used to connect devices to a network and transfer data.

→ Monitor :- A monitor is an output device that display the rasberry Pi's screen, information, Images and Videos.

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-> Keyboard :- A keyboard is an input device used to center text, number & commands into the Rasberry Pi

-> Mouse :- A mouse is an input device used to move the pointers select, items open file & interact with rashbegger

-> USA Derka :- USB (Universal storal Bus) devices are device connected to the sasberry pi through USB ports, such as open drive, keyboard, mouse or others peripherals.

-> Steps.

Step 1: Insert the Micro SD card.
Step 2: Connect the Monitor
Step 3: Connect the keyboard.
Step 4: Connect the Mouse
Step 5: Connect Ethernet
Step 6: Conflict a USB device.
Step 7: Power on the rasberry Pi
Step 8: Check the devices.

Practical No. 3

Aim :- Performs a practical to configure rasberry Pi Gp To pins for hardware interfacing.

-> Requirement

- Rasberry Pi
- Rasberry Pi OS
- Monitor

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- Keyboard
- Mouse
- Power supply
- Bread board
- LED
- Resister
- Lump wires.

→ Steps.

Step 1: Start the rasberry Pi
Step 2: Open the Terminal
Step 3: Check the GPIO interface.
Step 4: Identify the GPIO pin.
Step 5: Connect the hardware.
Step 6: Configure the GPIO pins as an output.
Step 7: Send a signal through the GPIO pin.
Step 8: Run the program.
Step 9: Stop the clean up.

* Rasberry pi pinout :-

1] 3V3 Power :- It provides a constant 3.3-volt power supply to external components connected to the Rasberry Pi

2] 5V power pin :- It provides a constant 5-volt power supply to external component connected to the rasberry Pi

3] Ground (GND) : It provides a common reference

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point for electrical circuit & complete the flow of current.

4] 1-WIRE :- Used to communicate with 1-wire devices such as temperatures sensor using single data wire.

5] DPI (Display parallel interface) used to connect rasberry Pi to parallel RGB display.

6] GPCLK (General Purpose Clock) :- General purpose clock pins can be setup to output a fixed frequency without any on going software control.

7] I2C (Inter integrated circuit) :- GP102 & GP103 the Rasberry Pi's I2C pins. allows for two-wire communication with a variety of external sensor and devices.

8] JTAG (Joint test Action Group) :- JTAG is a standardized interface for delnigging intergrated circuit which you can use to delnig your rasberry Pi

9] PCM (Pulse-code modulation) :- PCM is a digital form of analog audio used by rasberry Pi to send high quality sound to a DAC.

10] PWM (Pulse with modulation) :- PWM create a rectangular waves speed.

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signal used to control LED brightness display backlight an metor

11] SDIO (SD card interface) :- SDIO is the SD host/ MMC interface on the rasberry Pi SD signals are normally used for the micro SD slot.

12] SPI (Send peripheral interface) :- SPI is a joy-wire serial used to connect multiple devices using-chip-select ppns SPID :- GPIO - 1-11, SPI1 : GPID 16-21

13] UART (Universal psynchranas Receiver) :- Transmitter used to serial communication Rasberry Pi and sentor display and other devices.

14] wiring-Pi :- Wiring Pi and attempt to bring Artuine wiring like simplicity to the rasberry Pi

Practical No 4.

Aims :- Perform a practical to interface LED's with rasberry Pi GPIs pins

--> Requirements.

- Rasberry Pi board.
- LED
- 220 Ω resister
- Bread board
- Jymper wires.
- Power adapter.

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- Monitor
- HDMI cable
- Keyboard
- Mouse
- Rasberry Pi OS.
- Internet connection

→ Steps.

Step 1:- Connect the Rasberry Pi to the monitor keyboard mouse and power supply.

Step 2:- Place the LED on bread board & identify I is a node (+) and cathode (-) Terminals.

Step 3:- Connect a 120Ω resister in a series with the LED.

Step 4:- Connect GPIO. It (physical pins R) of the rasberry pi to the resistor.

Step 5:- Connect the cathode (-) of the LED to GND (physical pin ( ) of the rasberry Pi

Step 6:- Connect the other end of the resistor to the anode (+) of the LED.

Step 7:- Open the terminal

Step 8:- Write the python program.

Step 9:- Save the program

Step 10:- Run the program

Step 11:- Observe the LED.

Step 12:- Stop the program.

Code :-

```
import Rpi.GPIO is GPIO
import time
```

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```
# Use BCM GPIO numbering LED_PIN_1
# Set up GPIO
GPIO.setmode(GPIO.BCM)
GPIO.setup(LED_PIN, GPIO.OUT)
Print("Blinking LED. press ctrl+c to stop")

try:
while True:
GPIO.output(LED_PIN, GPIO.HIGH)
# Turn LED on time.sleep(1)
# Wait 1 second
GPIO.output(LED_PIN, GPIO.LOW)
# Turn LED off time.sleep(1)
# Wait 1 second
except keyboard interrupt:
print("Program stopped by users":)

Finally
GPIO.cleanup()
# Reset GPIO settings
```

Practical No. 5

Aims:- Performs a practical to display different LED patterns using Raspberry Pi GPIO programming

```
import RPi.GPIO as GPIO
import time.

# Use BCM GPIO numbering
LED_PIN 1 = 7
```

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```
LED-PIN2 = 17

# setup GPIO
GPIO.setmode (GPIO.BCM)
GPIO.setup (LED-PIN1,
GPIO.OUT)
GPIO.setup (LED-PIN 2,
GPIO.OUT)
print ("Blinking Two LED.Press ctrl+C to stop")
try:
while True:
    GPIO.Output (LED-PIN 1,
GPIO.HIGH)
    GPIO.output (LED-PIN 2,
GPIO.LOW)
    time.sleep (1)
GPIO.output (LED-PIN 1,
GPIO.LOW)
    GPIO.output (LED-PIN 2,
GPIO.HIGH)
    time.sleep (1)
except keyboard Interrupt:
print ("Program Stopped by user".)
finally:
GPIO.cleanup () # Reset
GPIO settings.
```